

Akash Shanmugam

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Education

The University of North Carolina at Charlotte

Charlotte | Dec 2024

B.S. in Computer Science - Data Science Concentration, Minor in Math

Coursework: Intro to Machine Learning, Cloud Computing, Data Mining, Software Engineering, Object-Oriented, Data Structures, OS and Networks, Linear Algebra, Probability & Statistics, Calculus

Skills

Languages: Python, HTML/CSS, Java, C, C++, C#, and SQL

Tools: Git/GitHub, Unix/Linux, Anaconda (Pandas/Scikit), AWS/Hadoop, Flask, Figma/Bootstrap, Node.js

Experience

Software Engineering

Jan-May 2024

- Regularly used Figma to plan UI design for several websites while integrating the prototype into a final design while integrating CSS for a better user experience. For the backend, SQL database integration with python repo connection was utilized.
- This experience involved being a product manager working in a team for multiple projects using Agile and Git while coordinating weekly tasks to run an effective project timeline and focused goal.

Cloud Computing Data Management

Aug-Dec 2023

- Completed a training course in AWS and used Hadoop cloud computing to manage and modify data tables on page rank and analyze Climate data on Asheville NC.
- This experience helped impact understanding of how climate rainfall data can be used to predict local effects of hurricanes while showing the benefit of cloud platforms of data analysis.

AI creation for integration in games

Jan-May 2024

- Created AI algorithm scripts for enemy "agents" in Unity for game development such as Chess and a 2-player bomber belt game (C#).
- The project used the base concept of how AI can be used to achieve certain end states using a variety of completely different game frameworks.

Projects

Full-Scale Event Planner Web Application Presentation

Apr 2024

- Created an Event Planner app that involved several Google API (maps, login, auto-email, etc.) features using web development knowledge of Python, Flask, SQL and HTML which was planned using an ERD and Agile development put online using Openrender.
- The project used github project guidelines and was presented in front of a panel of software engineers over the course of 20 minutes for review and critique.

Data Visualizations using Python Data libraries

Sep 2024

- Used Python to apply to visualization and prediction in Data Models with Anaconda/JupyterNotebook in a presented paper (Pandas, Matplotlib, Seaborn, Scikit-learn). Datasets used for different projects include Pokemon moves and Bat population distribution.
- Examples of models used were KNN and Logistic Regression using training data that was cleaned and tested multiple times.

Scalable database login management

Dec 2023

- Developed SQL queries in both Datagrip Postgres and MySQL for use in websites that can store users, hash passwords, and update and delete entries while handling scalability for multiple users.
- This format of login management was used for success in multiple web application projects.

Previous Education/Extracurriculars

US Citizen

UNCC 49th Security Division Club Member

Charlotte | 2022-2024

High School Diploma; Apex High School, Apex, NC

Cary, NC | 2020

- Robotics Team – Programming, Engineering, and Secretary

Tutor - AP Mathematics courses

2024